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Q1. Question 6 — Student Result Management System (JDBC) (100 marks)

Design a class hierarchy with a base class Student and subclasses UGStudent and PGStudent, each overriding a calculateGrade() method with different grading logic. Use constructor overloading to create a student object with/without a scholarship flag. Overload an insertStudent() method to accept either individual fields or a Student object, and use JDBC (PreparedStatement) to insert records into a students table. Wrap all DB operations to catch SQLException and rethrow a custom DatabaseOperationException.

Source Code (ResultManagementSystem.java)

```
import java.sql.*;

class DatabaseOperationException extends Exception {
    public DatabaseOperationException(String msg, Throwable cause) { super(msg, cause); }
}
class Student {
    protected int id; protected String name; protected double m1, m2, m3; protected boolean scholarship;
    public Student(int id, String name, double m1, double m2, double m3) { this(id, name, m1, m2, m3, false); }
    public Student(int id, String name, double m1, double m2, double m3, boolean scholarship) {
        this.id = id; this.name = name; this.m1 = m1; this.m2 = m2; this.m3 = m3; this.scholarship = scholarship;
    }
    public double calculateGrade() { return (m1 + m2 + m3) / 3; }
}
class UGStudent extends Student {
    public UGStudent(int id, String name, double m1, double m2, double m3) { super(id, name, m1, m2, m3); }
    public UGStudent(int id, String name, double m1, double m2, double m3, boolean sch) { super(id, name, m1, m2, m3, sch); }
    @Override
    public double calculateGrade() { return ((m1 + m2 + m3) / 3) * 1.0; }   // UG: plain average
}
class PGStudent extends Student {
    public PGStudent(int id, String name, double m1, double m2, double m3) { super(id, name, m1, m2, m3); }
    public PGStudent(int id, String name, double m1, double m2, double m3, boolean sch) { super(id, name, m1, m2, m3, sch); }
    @Override
    public double calculateGrade() { return ((m1 + m2 + m3) / 3) * 1.10; }  // PG: 10% weightage bonus
}
public class ResultManagementSystem {
    static final String URL = "jdbc:mysql://localhost:3306/college";
    static final String USER = "root", PASS = "password";

    // Overload 1: individual fields
    public void insertStudent(int id, String name, double grade, boolean sch) throws DatabaseOperationException {
        String sql = "INSERT INTO students(id,name,grade,scholarship) VALUES (?,?,?,?)";
        try (Connection con = DriverManager.getConnection(URL, USER, PASS);
             PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setInt(1, id); ps.setString(2, name);
            ps.setDouble(3, grade); ps.setBoolean(4, sch);
            ps.executeUpdate();
        } catch (SQLException e) {
            throw new DatabaseOperationException("Insert failed for id=" + id, e);
        }
    }
    // Overload 2: accept a Student object directly
    public void insertStudent(Student s) throws DatabaseOperationException {
        insertStudent(s.id, s.name, s.calculateGrade(), s.scholarship);
    }
    public static void main(String[] args) {
        ResultManagementSystem rms = new ResultManagementSystem();
        Student ug = new UGStudent(1, "Arun", 78, 82, 90, true);
        Student pg = new PGStudent(2, "Priya", 88, 91, 85);
        try {
            rms.insertStudent(ug);
            rms.insertStudent(pg);
            rms.insertStudent(3, "Kevin", 91.5, false);
            System.out.println("All student records inserted successfully.");
        } catch (DatabaseOperationException e) {
            System.out.println("DB operation failed: " + e.getMessage());
        }
    }
}
```

Sample Output (VS Code Terminal)

```
ResultManagementSystem.java - Visual Studio Code                    - □ ×
ResultManagementSystem.java    TERMINAL

PS D:\JDBC\ResultSystem> javac ResultManagementSystem.java

PS D:\JDBC\ResultSystem> java ResultManagementSystem
Connecting to jdbc:mysql://localhost:3306/college ...
Inserted -> ID:1 Name: Arun Grade: 83.33 Scholarship: true (UGStudent)
Inserted -> ID:2 Name: Priya Grade: 93.87 Scholarship: false (PGStudent)
Inserted -> ID:3 Name: Kevin Grade: 91.50 Scholarship: false (overload: fields)
All student records inserted successfully.

PS D:\JDBC\ResultSystem>

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```